

A refinement of the $7/8$ zero-free boundary for the Riemann zeta function

Leo Wei

October 8, 2026

Abstract

We refine OpenAI's family-003 zero-free theorem to the boundary $B = 4374785349/5000000000 = 0.8749570698$ for its finite-order Hecke family over $\mathbb{Q}(\sqrt{-3})$, and hence for Dirichlet L -functions and the Riemann zeta function. The argument varies the total prime-slot length and the aspect ratio of the compensated probe, retains an explicit reflected-energy loss, and transports the Euler correction and contour comparison to the new boundary. It uses the existing fourth-moment theorem at $\kappa = 3/4$. A complete Lean formalization includes the original $7/8$ baseline, the actual low and high bounds for one normalized probe, and the final nonvanishing statements. An exact two-bin obstruction leaves less than 10^{-12} of possible improvement within the specified two-parameter counting estimate. This restricted optimality result does not cover other counting estimates or probe constructions.

1 Introduction and main result

OpenAI's manuscript 003 proves a zero-free half-plane with boundary $7/8$ for a family of Hecke L -functions and transfers it to the Dirichlet and Riemann zeta functions [003]. We retain its arithmetic and moment estimates and change the parameters in its compensated-probe argument. The resulting improvement is

$$\frac{7}{8} - B = \frac{214651}{5000000000} = 0.0000429302.$$

Let $F = \mathbb{Q}(\sqrt{-3})$, and let $\mathfrak{X}_F$ denote the full finite-order Hecke family used in 003, including imprimitive characters. We use the same ideal coefficients and L -functions. To agree exactly with the formalization, define

$$\beta_* = \sup \left(\left\{ \frac{1}{2} \right\} \cup \left\{ \operatorname{Re} \rho : \begin{array}{ll} \chi \in \mathfrak{X}_F, & \operatorname{Re} \rho > 0, \\ L_F(\rho, \chi) = 0, & \rho \text{ is not a pole} \end{array} \right\} \right). \quad (1)$$

The inserted value $1/2$ makes the set nonempty. The Euler product bounds it above by 1, and the primitive-reduction theorem of 003 supplies primitive zero witnesses arbitrarily close to β_* whenever $\beta_* > 1/2$.

Theorem 1. *For the family in (1),*

$$\beta_* \leq B := \frac{4374785349}{5000000000} = 0.8749570698.$$

Consequently, $L_F(s, \chi) \neq 0$ for every $\chi \in \mathfrak{X}_F$ with $\operatorname{Re} s > B$, away from its principal pole. For every integer $q \geq 1$ and every Dirichlet character χ modulo q ,

$$L(s, \chi) \neq 0 \quad (\operatorname{Re} s > B, \quad \neg(\chi \text{ principal and } s = 1)).$$

In particular, $\zeta(s) \neq 0$ when $\operatorname{Re} s > B$ and $s \neq 1$. No assertion is made about zeros on $\operatorname{Re} s = B$.

The proof starts from the established ceiling $\beta_* \leq 7/8$ in 003. Under the contrary assumption $\beta_* > B$, we construct a single normalized physical probe and compare it with the same principal Mellin signal on both sides of the argument. The essential changes are an explicit loss in the reflected-row estimate, a change in the principal signal exponent, and a new exact row-count certificate. All estimates for the changed geometry are proved below or imported with their original hypotheses checked. No extension of the source's fourth-moment range is required.

The companion Lean development proves Theorem 1, including its source baseline, from the actual analytic and arithmetic objects. The final declarations have no assumed probe estimates, moment bounds, or zero-free ceiling. Section 8 gives the proof entry points and verification details. The construction is asymptotic; it does not provide a practical height threshold or an iteration toward the Riemann hypothesis.

Source inputs and notation

We use the version of 003 pinned to commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. The notation [003, TeX lines a–b] refers to lines *a–b* of its unchanged `build/paper.tex`; the bibliography gives the permanent link. The source inputs, in addition to its family ceiling, are:

1. the arbitrary-positive-scale base-probe identities and finite local replacement formulas [003, TeX lines 3470–3478, 4010–4198, 8688–8795];
2. the reflected-energy estimate with its row and powerful-part constraints, and the quantitative Gram estimate [003, TeX lines 7747–7805, 8341–8358];
3. the buffered detector, which requires $\beta_* > 51/100$, and its witness-height and original-zero-mask conditions [003, TeX lines 4213–4669];
4. the marked inverse moment, sixth-power amplification, and plain fourth moment with their strict widths and coefficient classes [003, TeX lines 9250–9299, 12362–12514, 12531–12624];
5. the fixed-ray prime normalizer and the smooth-separation, Sobolev and Mellin-tail estimates, with their seminorm dependence;
6. the variable-boundary continuation criterion and family transfer [003, TeX lines 401–501, 6707–6777].

The source's fixed-geometry low estimate, final row-count application, and principal contour interfaces are replaced by the arguments below. In particular, the source's ambient contradiction $\beta_* > 7/8$ is never used in the new contradiction regime $B < \beta_* \leq 7/8$.

For an integral ideal or its prescribed primary generator n , write $q_n = N(n)$ for its absolute norm. The finite set S contains the fixed excluded primes. The function $L_F^S(s, \eta)$ is obtained by deleting the Euler factors at S from $L_F(s, \eta)$, equivalently by using the source's excluded-prime character. All ray restrictions, completed-index masks and zero extensions are those of 003. Implicit constants may depend on fixed arithmetic data, windows and the target character; positive exponent savings identified as uniform do not depend on the target.

2 Geometry and the common physical expression

Put $p = 214651/1250000000$ and $b = 192821/1562500$, and set

$$\begin{aligned} \ell &= \frac{1}{6} + p, \quad l_x = \frac{1 - \ell - b}{2}, \quad l_y = \frac{1 - \ell + b}{2}, \\ h &= \frac{1 + 3\ell + b}{2}, \quad M = l_x + l_y = \frac{5}{6} - p, \quad X = Z^{l_x}, \quad Y = Z^{l_y}. \end{aligned} \quad (2)$$

Thus $M + \ell = 1$, $h = 1 - l_x + \ell$, and

$$C(s) = s + \frac{l_x}{2} - 1 + \frac{h}{6} = s - \frac{2}{3} - \frac{b}{6}, \quad \frac{l_x}{2} + \frac{b}{12} = C(B) = \frac{938614149}{5000000000}. \quad (3)$$

Fix finitely many positive slot lengths ℓ_i summing to ℓ , with $P_i = Z^{\ell_i}$, and nonnegative nonzero smooth annular weights W_i whose underlying prime windows are disjoint. Their arithmetic restrictions are exactly those in 003. Write $p_J = \prod_{i \in J} p_i$. More precisely, the allowed prime set for slot i is

$$\mathcal{P}_i(Z) = \{p \text{ prime} : p \notin S, p \in 1_T, q_p/P_i \in \text{supp } W_i\},$$

where 1_T is the fixed principal ray class in the source. Fix also two nonnegative, nonzero smooth base windows W_0, W_1 with compact support in $(0, \infty)$. These are the windows in the source's base probe and determine its positive principal residue constant. The construction below may take a common smooth window supported in $(1, 2)$ for both base windows and all slot weights. The physical expression is

$$\begin{aligned} I_\eta^{\text{mod}}(Z) &= \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J \subseteq \{1, \dots, K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} \\ &\quad \cdot I_{\eta; p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \end{aligned} \quad (4)$$

Here $I_{\eta; D}$ is the source's base probe with D dividing its completed index; see [003, TeX lines 6894–6913]. Every slot window remains at its original scale in every summand. In particular, marked tuples will be summed inside the completed-row factor, while rescaled tuples are counted separately. The same expression is used on both sides of the argument.

Here is the exact high identity needed for that assertion. The base identity holds for arbitrary positive X, Y, Z . In the source's local notation, at a selected prime of norm Q the two-term operation replaces the factor P_p by

$$\overline{\eta(p)} Q^{s+z-1} P_p^* - Q^{z-w-1} P_p = \frac{1-D}{(1-V)(1-W)} Q^{z-1} G_p.$$

This algebraic identity extracts precisely the same scalar local factor as at an unselected prime and involves no division by P_p or H_p . Finite composition over the disjoint slots therefore gives the source's exact compensated high integral [003, TeX lines 8775–8795], with the perturbed values of X, Y, Z and the same full correction $\mathfrak{H}_{\eta, u, Z}$. There is no additional scalar factor. Normal convergence in the enlarged domain is proved in Lemma 3.

3 Transport of the direct estimate

Lemma 2. *For every fixed admissible finite slot system in (4) and every $\epsilon > 0$,*

$$|I_\eta^{\text{mod}}(Z)| \ll_{\eta, \epsilon, \text{data}} Z^{C(B)+\epsilon}.$$

The exponent contains no slot-mesh loss. Constants and lower thresholds may depend on the fixed slot system.

Proof. For a rescaled subset let $d = \sum_{i \in J} \ell_i$, $M' = M - 2d$, $\ell' = \ell - d$, $X' = X/q_{pJ}$, and $Y' = Y/q_{pJ}$. The annular ratio $r_J = q_{pJ}/Z^d$ lies in a fixed compact subinterval of $(0, \infty)$. The source's exact low separation has prefactor $Q^{-1/2}$, where $Q = q_{b*} X' Y'$, and factors $A_m(Y')$ and $B_m^J(Z)$; all fixed ray sums and the external smooth height integral are retained [003, TeX lines 3470–3478, 8093–8106].

We first replace the fixed-geometry completed-row bound by

$$\sum_{0 < q_m \ll Q} |B_m^J(Z)|^2 \ll Z^{M' + \lambda(d) + \epsilon}, \quad \lambda(d) = \frac{(5\ell - 1 + d)_+}{4}. \quad (5)$$

To derive it, apply the general reflected-energy lemma [003, TeX lines 7747–7805] to the source's actual residual-row dyads. Use its notation $O, H, A_0, N_0, B_0, S_0, z_a, v, \ell_b, e_\lambda$, and write $\Delta_H = M' - O - H \geq -O(1/\log Z)$. Its powerful-part constraints give $2A_0 \leq O + O(1/\log Z)$, $N_0 \leq A_0$, and $z_a \leq \ell'$. For a Gaussian annulus, the completed length is $N_* = 1 + \ell' + \theta_N$, with θ_N arbitrarily small after the prescribed annular truncation. Its available dual length is

$$T_d = 2H + 2A_0 + 2z_a - 1 - \ell' - \theta_N - N_0 - 3B_0.$$

The exact identity $M' + \ell' - 1 = -3d$ implies $T_d \leq H - 3d + O(\epsilon)$ with the usual fixed-data threshold. Retention therefore yields $\max(H, v + \ell_b) = H + O(\epsilon)$. The kernel saving is extracted at this actual H before squaring; no shorter row added by a positive enlargement changes its scale.

Let $s_0 = \min\{v, z_a, (v + z_a)/3\}$. If $s_0 = z_a$, the reflected energy is at most $M' - O/2 - \Delta_H + O(\epsilon)$. Otherwise $s_0 \geq v/2 - O(1/\log Z)$. With $y = v + 3\ell_b + e_\lambda$,

$$s_0 + \ell_b + 2e_\lambda/3 + (T_d - y)_+/2 \geq T_d/4 - O(\epsilon).$$

The charge identity in [003, TeX lines 8300–8308] is

$$\begin{aligned} O/2 + \Delta_H + S_0 + B_0 + T_d/4 &= \frac{2M' + 2z_a - 1 - \ell' + 2\Delta_H - \theta_N}{4} \\ &\quad + \frac{2A_0 - N_0 + B_0 + 4S_0}{4}. \end{aligned}$$

The second numerator is nonnegative. Keeping the remaining $+z_a$ and using $z_a \leq \ell'$ bounds this second branch by $M' + (1 + 3\ell' - 2M' - 2\Delta_H)/4 + O(\epsilon)$. Taking the maximum of the two branches and using $1 + 3\ell' - 2M' = 5\ell - 1 + d$ proves (5). The source's joint-profile separation, whole-annulus Gaussian tails, and retained-dyad kernel tails apply on the same bounded length ranges. All masks and product-form coefficients are unchanged.

For the quantitative Gram estimate [003, TeX lines 8341–8358], $P_a = Y'^2/Q = q_{b*}^{-1} Z^b$ is independent of J . The needed positive margins include

$$M' \geq \frac{624356047}{1250000000} > \frac{49}{100}, \quad l_x - \ell > \frac{18}{100}, \quad l_y - \ell - \frac{11b}{6} > \frac{8}{100}.$$

Hence its second Gram term is absorbed as in the source, and

$$\sum_m |A_m(Y')|^2 \ll (1 + |v|)^{J_{\text{Gr}}} (Q/Y') P_a^{1/6} Z^\epsilon.$$

Cauchy–Schwarz with the exact $Q^{-1/2}$ prefactor gives $\ll (X')^{1/2} P_a^{1/12} Z^{\lambda(d)/2 + \epsilon}$, up to the bounded annular ratio and fixed constants. The external scalar $q_{pJ}^{-3/2}$ and $(X')^{1/2}$ together contribute q_{pJ}^{-2} . On slot i , the compact positive support implies $q_p \geq a_i Z^{\ell_i}$ for a fixed $a_i > 0$. Therefore

$$\sum_p |W_i(q_p/P_i)| q_p^{-2} \ll Z^{-\ell_i/2} \sum_p q_p^{-3/2} \ll Z^{-\ell_i/2}.$$

The last convergent mass is a special case of the source's supported-ideal summability estimate. Multiplication over the finite rescaled subset gives $Z^{-d/2}$; consequently its total extra exponent is

$$-d/2 + \lambda(d)/2 = -d/2 + (5\ell - 1 + d)_+/8 \leq -3d/8 \leq 0.$$

Here $\ell < 1/5$ gives $(5\ell - 1 + d)_+ \leq d$. The finite supported-mass inequalities are checked in `Transport/SupportedMass.lean`. The actual supported arithmetic decay and its combination with reflected energy and the Gram estimate are checked in `PhysicalProbe/SupportedDecay.lean` and `PhysicalLow/LowCentralTuple.lean`. The finite subset sum, logarithmic dyads, and external smooth integral cost arbitrarily small powers for fixed K . Equation (3) finishes the proof. $\square$

4 Local analytic extension and normalization

Set $r = p/4 = 214651/5000000000$ and

$$\mathcal{D}_2(B) = \{s_r \geq B, w_r \geq 19/20, z_r \geq 33/200\}.$$

The first Euler region of 003 is left unchanged.

Lemma 3. *The exact local Euler correction and full selected-tuple correction extend holomorphically to neighborhoods of points of $\mathcal{D}_2(B)$, with the source's required all-height bounds. For a sufficiently large fixed exclusion cutoff,*

$$\sup_{\operatorname{Re} s > B} |H_\eta(s) - 1| \leq \frac{1}{2}, \quad H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6).$$

At a selected principal prime, $\mathcal{B}_p = -1 + O(q_p^{-B})$, uniformly in all imaginary parts.

Proof. Use the exact rational factors and defect identity in [003, TeX lines 4010–4198]. On the enlarged region their denominators satisfy

$$|R| \leq Q^{-56/25+6r}, \quad |V| \leq Q^{-99/100}, \quad |D| \leq Q^{-B} < 1.$$

These are uniformly separated from one. At a good prime the terms DV, DW, DVW, VW and the grouped $\mathcal{E}_p$ terms have respective powers

$$-s_r - 6z_r, \quad -s_r - w_r, \quad -s_r - w_r - 6z_r, \quad -6z_r - w_r, \quad 4 - 6s_r - 6z_r, \quad 1 - s_r - w_r - 6z_r.$$

All decrease in the real variables. Their maximum at the corner is $-363/200 + r < -1$. At a ramified prime the common term, strict family, and boundary terms have powers bounded by

$$\begin{aligned} 4 - 6s_r - 6z_r, \quad 1 - s_r - w_r, \quad -s_r - w_r, \quad 3/2 - 3s_r, \\ 2 - 3s_r - w_r, \quad 2 - 4s_r, \quad 5/2 - 4s_r - w_r, \quad 3 - 6s_r. \end{aligned}$$

Their maximum is $-33/40 + r < 0$. Thus

$$H_p - 1 = O(Q^{-363/200+r}) \ (p \nmid u), \quad H_p - 1 = O(Q^{-33/40+r}) \ (p \mid u).$$

Normal convergence and the divisor-product bound give $\mathcal{H}_{\eta,u} \ll_\epsilon q_u^\epsilon$. For $u = 1$, the good-prime tail above P_0 is $O(P_0^{-163/200+r})$, whose decay exponent exceeds $4/5$. A sufficiently large fixed cutoff gives the asserted uniform contraction. Further fixed exclusions preserve its positive majorant.

The full correction is the finite tuple sum using the holomorphic replacements G_p [003, TeX lines 8688–8755]; it is not globally defined by division by H_p or P_p . Its unselected products converge normally. On a fixed real box, each selected factor is $O(Q^c)$ for a fixed c , so the tuple sum is $O(q_u^\epsilon Z^{c'})$, uniformly in heights, with c' fixed before any external tail order.

For the principal row only, the cutoff makes H_p nonzero. The four error powers for $G_p/H_p + 1$ are $-s_r, -6z_r, 4 - 5s_r - 6z_r, 1 - w_r - 6z_r$, each at most $-B$. This proves the last assertion and the principal tuple bound $|\mathfrak{H}_{\eta,1,Z}| \ll Z^{\ell z_r}$. $\square$

Define $S_i(Z) = \sum_{p \in \mathcal{P}_i} W_i(q_p/P_i) q_p^{-5/6}$ and

$$A_T(Z) = (-1)^K Z^{-\ell/6} \prod_i S_i(Z).$$

The source's fixed-ray prime normalizer implies $|A_T(Z)| \asymp (\log Z)^{-K} \neq 0$ for sufficiently large real Z . Positive windows and Lemma 3 give, for $0 < \kappa_P < B \min_i \ell_i$,

$$\mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) Z^{\ell/6} A_T(Z) (1 + O(Z^{-\kappa_P})) \quad (6)$$

on the entire line $\operatorname{Re} s = \beta_* + e$. Let $c_S > 0$ be the source's principal residue constant and put $J_\eta = I_\eta^{\text{mod}}/(c_S A_T)$. This is the normalization on both sides of the argument. Since K is fixed, $|A_T(Z)|^{-1} \ll_\epsilon Z^\epsilon$ for every $\epsilon > 0$. Applying Lemma 2 with half the requested loss gives the same low exponent for J_η , with any prescribed positive final loss.

5 Fixed-moment row counts and all frequency ranges

Suppose $B < \beta_* \leq 7/8$, and write $\Delta = \beta_* - B > 0$. The detector requires only $\beta_* > 51/100$. Each occupied nonfloor bin has an actual zero witness giving $a \leq \beta_*$, hence $\delta = 2a - 1 \leq 3/4$; the floor bin $a = 51/100$ is treated separately. Use the already allowed $\kappa = 3/4$ in the fourth moment: its hypothesis is $\beta_* \leq (1 + \kappa)/2$, supplied by the original $7/8$ theorem. The effective detector width is one in base $U = Z^d$, not the direct width M . The inverse amplification parameter remains $\alpha = 5/6$.

For nonfloor bins $a > 51/100$, the proof of the selected counts at [003, TeX lines 15185–15445] now uses

$$z_P(m) = \frac{2(1 - 2m)}{9}, \quad z_M(t') = \frac{1 - t'}{2}.$$

There is no dynamic-capacity surcharge. Let $g_i \in [0, \delta/2]$ be the source's logarithmic amplitude bin for slot i , assigning $g_i = 0$ to every error slot and every zero-amplitude slot. Define $q = \ell^{-1} \sum_i \ell_i g_i$ over all slots, including those error slots. Thus the amplitude ratio is $x = q/\delta \in [0, 1/2]$. Put

$$\begin{aligned} D_x &= 3 - 17x/9, & P_x &= (2 - 8x/9)(1 - x), \\ \mathcal{J} &= (\alpha - \delta)D_x + \delta P_x, & t_{\text{cut}} &= 1 + \frac{\delta P_x}{2\mathcal{J}}, \\ R_* &= 1 - \delta + (\alpha - \delta)(t_{\text{cut}} - 1). \end{aligned}$$

The source proof specializes to $\#\mathcal{B} \ll U^{R_* + \epsilon} (1 + T_1)^A$. For the positive-slot hypothesis, a sufficiently large physical sixth-power-free row coprime to S has a good prime with valuation between one and five. Its row character ramifies there, whereas every member of the fixed exceptional family Θ is unramified there. Thus moderate rows avoid Θ ; bounded unit rows are treated separately. The floor bin has no required witness and uses $R = 1$. Its inverse strict-width margin, zero-capacity neighborhoods, whole product conjugation, fixed character class, original zero masks, and Sobolev

treatment of witness heights are unchanged. In particular the source crossing supplies requested inverse length at most $7/37$ and second width margin at least $9/37$. This is a specialization of the proof at fixed κ ; the original application with positive $\beta_* - 7/8$ is not cited outside its stated hypotheses.

The general accounting identity gives the exponent relative to $C(B)$:

$$E_B(d) = a - B + h(17/50 - 1/6) - al_y - \ell/2 + q\ell + d(R + \delta/2 - 17/50). \quad (7)$$

For the balanced endpoint, $1 - \delta \leq R_* \leq 1$. We evaluate (7) directly using the full rational count, rather than its coarser original perturbation bound. The continuous certificate is

$$E_B(h) \leq -c_{\text{end}}, \quad c_{\text{end}} = \frac{1}{10^{12}}, \quad (\delta, x) \in [1/50, 3/4] \times [0, 1/2]. \quad (8)$$

Here is the exact verification procedure. The denominator $\mathcal{J}$ is positive since $\alpha - \delta \geq 1/12$, $D_x \geq 37/18$ and $P_x \geq 7/9$. Clearing that denominator produces the polynomial $Q(\delta, x) = \mathcal{J}(-E_B(h) - c_{\text{end}})$ of bidegree $(2, 3)$. An exact partition into 40 rational rectangles gives strictly positive Bernstein coefficients for Q on each rectangle. After rescaling a rectangle to $[0, 1]^2$, if $Q = \sum a_{ij}u^i v^j$, these coefficients are

$$c_{kl} = \sum_{i \leq k, j \leq l} a_{ij} \frac{\binom{k}{i} \binom{l}{j}}{\binom{2}{i} \binom{3}{j}}, \quad 0 \leq k \leq 2, \quad 0 \leq l \leq 3.$$

The basis functions are nonnegative. The companion `Transport/TunedEndpoint.lean` checks the polynomial identity, every rectangle certificate, and full domain coverage. Its theorem `adaptive_endpoint_bound` proves (8) without sampling assumptions. The exact rational rectangles and their coefficient checks are included in that Lean source; their verification does not require an optimization script. The remaining error-free comparisons are:

Range and input	Retained saving
Floor $a = 51/100$, $R = 1$	$34448969/5859375000 > 1/200$
$1/100 \leq d \leq 1/2$, $R = 76/75 - 2\delta/3$	$36605252263/1500000000000 > 3/125$
$1/2 \leq d \leq h$, balanced count	c_{end}
Small nonprincipal rows $u \neq 1$, $q_u \leq Z^{1/100}$, relative to $C(\beta_*)$	$1/20$ (before the $8e$ error)

The moderate slopes are positive, so the stated endpoints control their full intervals. For $h < d \leq h + \zeta$ take $\zeta = c_{\text{end}}/8 = 1/(8 \cdot 10^{12})$; the slope cost is at most 2ζ , leaving

$$c_0 = c_{\text{end}} - 2\zeta = \frac{3}{4 \cdot 10^{12}} > 0. \quad (9)$$

Supply persists because

$$5\ell - h - \zeta = \frac{533559267197}{2400000000000} > 0, \quad \frac{\ell}{h + \zeta} > \frac{1}{5} > \frac{7}{37}.$$

The no-slot count treats the intermediate range. Bounded nonprincipal unit rows are part of the small-row estimate. Beyond $h + \zeta$, the unchanged absolute line $(s, w, z) = (2, 2, z_\infty)$ with fixed sufficiently large z_∞ supplies a saving larger than c_0 .

Extent of the optimization. The same affine geometry may be tested with arbitrary p, b while keeping the specified balanced count R_* and $\alpha = 5/6$. Two admissible bins, $x = 1/2$ and $\delta_1 = 38858371/10^8$, $\delta_2 = 38858372/10^8$, give an exact obstruction for every real b . Write their endpoint exponents as $E_i(p, b) = A_i + s_i p + t_i(b - 1/8)$, where $t_i = R_*(\delta_i, 1/2)/2 - 1/3$. Then $t_1 > 0 > t_2$. The positive weights $w_1 = -t_2/(t_1 - t_2)$ and $w_2 = t_1/(t_1 - t_2)$ sum to one and cancel b . Exact arithmetic gives

$$w_1 E_1 + w_2 E_2 = S(p - p_*), \quad S > 0, \quad p_* = \frac{4562748652437347}{26570739037370607802}.$$

Thus every $p \geq p_*$ fails the strict uniform endpoint inequality, regardless of b . For the chosen parameter,

$$0 < B - \left(\frac{7}{8} - \frac{p_*}{4} \right) < 10^{-12}.$$

The companion Lean certificate checks this obstruction as well. It is an exact necessary bound for this two-parameter endpoint estimate, not an assertion that the limiting value is attained. In particular, retuning p, b with the same balanced count cannot improve the chosen boundary by 10^{-12} . Sharper counts, restrictions on which bins occur, or different moment estimates or geometries are not excluded. The certificate makes no claim about the difficulty of such changes. The extremely small retained saving also requires fresh choices of losses and slot count; no practical bound on the resulting constants or height threshold is claimed.

6 Contour interfaces and noncircular choices

The shared interfaces in 003 literally start at $\sigma_0 = 7/8$; their proofs must be read with the following substitutions. Every signal shift $-11/16$ is replaced by $-2/3 - b/6$, as dictated by (3). The principal contours use $\mathcal{D}_2(B)$ and (6); their only crossed poles remain $w = 1, z = 1/6$. Their positive margins are $m_w = l_y/20$, $m_z = h/600$, and κ_P . Nonprincipal contours lie in the unchanged first Euler region with $\text{Re}(s + w) \geq 1 + 10e$. For small nonprincipal rows the formalized argument retains the source's original local domain: take $s_r = \max\{\beta_*, 7/8\} + 8e$, $w_r = 1/2$, and $z_r = 17/50$. This lies strictly to the right of β_* and satisfies $s_r \geq 7/8$. Since $\beta_* \geq B$, the extra cost obeys $\max\{\beta_*, 7/8\} - \beta_* \leq 7/8 - B$. The resulting physical scale is at most $Z^{C(\beta_*) - 7/100 + 8e}$; spending $2/100$ on the dyadic row sum gives $O_\eta(Z^{C(\beta_*) - 1/20 + 8e})$ before the normalization loss. No extension of the small-row local table below $7/8$ is needed. Global reciprocal control is invoked strictly to the right of β_* ; the final nonprincipal move is inside a buffered zero-free rectangle. No reciprocal estimate assumes zero-freeness to the right of B .

The parameter construction used by the Lean implementation has the following dependence order. Fix $t = 10^{-15}$, and let $\eta_{\text{mesh}} > 0$ be the plain-moment mesh supplied by the source at this loss and at the fixed moment parameter $\kappa = 3/4$. It is fixed before the slot count. Choose a finite positive even K and distinct slot lengths satisfying

$$\sum_i \ell_i = \ell, \quad \ell_{\min} := \frac{\ell}{2K} \leq \ell_i < \frac{2\ell}{K} \leq \min\{t/200, \eta_{\text{mesh}}/200\}.$$

The factor 200 accounts for the minimum conductor exponent $d \geq 1/200$. The construction in `PhysicalHigh/DistinctSlots.lean` permits arbitrarily large even K . Use the same nonnegative nonzero smooth window supported in $(1, 2)$ at every slot. Distinct exponents give disjoint prime scales for all sufficiently large Z . Set

$$d_{\max} = h + \zeta + 2t, \quad r_{\min} = \ell_{\min}/d_{\max}.$$

Thus $d_{\max} r_{\min} \leq \ell_i$, and the checked numerical conditions include $d_{\max} \leq 1$ and $(7/37)d_{\max} \leq \ell$.

After these choices, select the detector error e , detector width ε , phase allowance, positive detector cost $c_{\det}$, smooth-loss parameter ϵ , and positive saving σ . The post-slot construction in `PhysicalHigh/Parameters.lean` makes the K -dependent errors small enough while ensuring

$$\sigma \leq t/2, \quad e \leq \ell_{\min}/4, \quad \sigma \leq \ell_{\min}/4, \quad \sigma + e \leq B\ell_{\min}.$$

The moment mesh's internal contour displacement is distinct from e . Choose one finite exclusion set and one smooth window, consistently across all targets and both probe bounds, and then choose the arithmetic count constants at loss t . These constants precede the moment theorem's polynomial height degree J_{mom} . The moment producer supplies J_{mom} uniformly over target characters, with a target-dependent constant and eventual threshold; its estimate is uniform in the height bound and in the subsequently chosen detector height exponent.

Only now choose a single $\tau > 0$ satisfying

$$\tau < 1/400, \quad 4\tau < c_{\det}/200, \quad 2\tau(1 + J_{\text{mom}}) \leq t, \quad \tau(2 + 4\epsilon) < t.$$

The moment height is $Z^{2\tau}$, and $(1 + Z^{2\tau})^{J_{\text{mom}}} \ll Z^t$. The detector subsequently supplies a finite index bound n ; $(3n + 1)Z^\tau \leq Z^{2\tau}$ holds eventually and creates no dependence cycle. The high estimate books its two height costs separately: the direct height cost is $2\tau \leq t$, while the polynomial moment cost is t . With conductor margin t and count buffer $\mu = 2t$, its complete exponent budget is

$$(159\varepsilon + 9t) + 2t + (26e + (K + 8)\epsilon + t + t\ell) + (t + 2\tau + t) + t \leq 18t < c_0.$$

The other checked budgets are

$$\begin{aligned} \sigma + 8e + t/8 &\leq 1/20, & \sigma + t/8 &\leq 1/4000, \\ 26e + (K + 8)\epsilon + t + t\ell + t/8 + \sigma &\leq 1/200, & \sigma + t/8 + t/8 &\leq t. \end{aligned}$$

They supply the small-row, principal, floor, and central savings. External tail orders are chosen after τ ; constants and eventual thresholds may depend on the target. The resulting σ is common to all targets. The low bound is requested separately with loss $\Delta/2$. This is an asymptotic existence argument; it gives no practical numerical threshold for Z .

7 Continuation and conclusion

The principal signal after the same normalization is

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds.$$

The preceding steps give, for every primitive target,

$$|J_\eta(Z)| \ll_\eta Z^{C(B) + \Delta/2}, \quad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*) - \sigma}, \quad \sigma > 0.$$

Set

$$m = \min\{\Delta/2, \sigma\} > 0, \quad a = \max\{B + \Delta/2, \beta_* - \sigma\} = \beta_* - m.$$

Then $B < a < \beta_* \leq 7/8$. Since $C(s)$ is affine with slope one, the triangle inequality gives

$$|f_\eta(Z)| \ll_\eta Z^{C(a)}.$$

The source’s Mellin identity, local integrability and rapid decay at zero apply to this same signal. Its variable-boundary continuation criterion [003, TeX lines 402–501] continues the regularized reciprocal to $\operatorname{Re} s > a$. On that half-plane, Lemma 3 gives $|H_\eta(s)| \geq 1/2$, and the Gaussian multiplier never vanishes. The regularized identity rules out nonpole zeros of $L_F^S(s, \eta)$ there. Finite Euler-factor deletion preserves the relevant zeros in $\operatorname{Re} s > 0$, so the original primitive $L_F(s, \eta)$ has no such zero to the right of a either.

The number m is independent of the primitive target character, although its implied constants and lower thresholds may vary. The primitive-supremum theorem supplies a primitive character and a nonpole zero ρ with $\operatorname{Re} \rho > \beta_* - m = a$, contradicting the previous paragraph. This proves $\beta_* \leq B$ without assuming that the supremum is attained. Finally, the variable-boundary family transfer [003, TeX lines 6707–6777] gives the Dirichlet and zeta conclusions of Theorem 1.

8 Lean formalization and reproducibility

The companion project [Lean] verifies the complete transport: actual reflected-energy and supported-mass bounds, the Euler correction and principal contour comparison, canonical tails, all four arithmetic moment producers, and both estimates for the same normalized probe. The proof revision is `f010b7a3fa42b72039417029c30e357cdb4197b2`. Its 133 transport modules match the sources used for the completed local verification. All 2,926 OpenAI modules under `lean/OAI/` match the pinned upstream source manifest, with both SHA-256 and Git-blob hashes checked.

The main proof entry points under `lean/Transport/` are:

File or directory	Content
<code>OriginalBaseline.lean</code>	Original 7/8 theorem from 003
<code>TunedEndpoint.lean</code> , <code>TunedCandidate.lean</code>	Continuous endpoint bound and exact obstruction
<code>PhysicalLow/</code> , <code>PhysicalProbe/</code>	Actual low estimate
<code>PhysicalAnalytic/</code> , <code>PhysicalTail/</code>	Euler products, contours, tails and continuation
<code>PhysicalHigh/</code>	Actual moment estimates and parameter choices
<code>CommonProbe.lean</code>	Assembly for one normalized physical probe
<code>Nonvanishing.lean</code>	Final Hecke, Dirichlet and zeta statements

The final declarations are `improved_hecke_boundary`, `hecke_nonvanishing`, `dirichlet_nonvanishing`, and `zeta_nonvanishing`, in namespace `RH.Transport`. The first bounds the actual full-family supremum in (1); the remaining declarations state nonvanishing for the source Hecke functions and the standard Lean Dirichlet and zeta functions. Lean’s zeta function is totalized at its pole; Theorem 1 states the conventional meromorphic conclusion away from that pole.

A completed local audit checked all 1,269 transport theorem declarations, including 798 public declarations, and the exact final theorem types. Their transitive axiom dependencies contain only `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx` or additional axioms. The published sources contain the complete proofs and rational certificates; the exploratory scripts and local audit report are not part of the repository. Lean is pinned to version 4.34.1 and `mathlib` to commit `d13f23b723b8a846827a245b89c10fc7d3f11612`.

For a fresh checkout, retain the committed Lake lockfile, fetch its pinned dependencies, apply the two supplied compatibility patches once, and build all transport modules. From the repository root:

```
cd lean
lake env lean --version
```

```

for pkg in rellich-kondrachov PrimeNumberTheoremAnd; do
  git -C ".lake/packages/$pkg" apply \
    "$PWD/upstream-config/patches/$pkg-lean4341.patch"
done
lake exe cache get
lake build

```

The first Lake command materializes the pinned dependencies without updating the lockfile. The patches and their recorded revisions are in `upstream-config/`; warnings about local changes in those two packages are expected after patching. To inspect the final theorem types and their transitive axiom dependencies directly, run from the same directory:

```

lake env lean --stdin <<'LEAN'
import Transport.Nonvanishing
#check RH.Transport.improved_hecke_boundary
#check RH.Transport.hecke_nonvanishing
#check RH.Transport.dirichlet_nonvanishing
#check RH.Transport.zeta_nonvanishing
#print axioms RH.Transport.improved_hecke_boundary
#print axioms RH.Transport.hecke_nonvanishing
#print axioms RH.Transport.dirichlet_nonvanishing
#print axioms RH.Transport.zeta_nonvanishing
LEAN

```

References

- [003] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\text{Re } s > 7/8$* , manuscript dated September 30, 2026. Family 003 in `openai/math`; pinned source commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. [Pinned TeX source](#).
- [Lean] LW56e2, *Quasi-Riemann boundary refinement in Lean*, 2026. [Companion repository](#); [pinned proof revision](#) `f010b7a3fa42b72039417029c30e357cdb4197b2`.